DOE Genesis Mission Phase I Proposal Package

DE-FOA-0003612

DOE Title Page

Project title:	From improved Coupled Ocean–Ice States to Improved Water Prediction: A Generative AI Approach
Lead Applicant/Institution:	Los Alamos National Laboratory
Street address:	Bikini Atoll Road, Los Alamos, New Mexico 87545-0001
Lead PI:	Luke Van Roekel
Administrative POC:	Kevin John
RFA number:	DE-FOA-0003612
Primary challenge and focus area:	Predicting U.S. Water for Energy; Weeks to Years Prediction

Senior/Key Personnel for Lead Institution and All Partner Institutions

Institution	Senior/Key Personnel
LANL	Andrew Roberts, Katrina Bennet, Trevor Hillebrand, Olawale Ikuyajolu
LLNL	Peter Caldwell, Emily de Jong
HPE	Andrew Shao
CU	Ian Grooms

Summary Budget Information for Lead Institution and All Partner Institutions

Institution	Year 1 Budget (USD)
LANL, Lead institution	515,000
LLNL	125,000
Hewlett Packard Enterprise	90,000
University of Colorado Boulder	20,000
Total	750,000

1 Introduction

Forecasts of United States water availability on timescales of hours to weeks have been revolutionized by AI based systems. Accuracy of these predictions now meet, and in many cases, exceed performance of traditional physics-based models [1, 2, 3]. However, beyond a few weeks, forecast accuracy for AI models declines due to a combination of factors, including deficient initial conditions [4, 5] - a problem also experienced by physics-based models [6]. Because advance preparation is critical for avoiding electricity outages and costly damage to energy infrastructure, limited forecast skill over weeks to years timescales restricts DOE's ability to anticipate weather-driven disruptions. Improving predictability and quantifying forecast uncertainty are therefore essential for advancing DOE's mission of energy security, resilience, and prosperity. Given that numerous studies have shown the importance of ocean and ice processes for predictability over the United States on timescales of weeks to years [7, 8, 9, 10, 11] advancing single component emulators to increasing resolution is insufficient. **The goal of this project is to prove that more accurate initialization of the coupled ocean - sea ice state through the use of state-of-the-art satellite-based sensors and generative AI methods will greatly improve weeks-to-years U.S. water predictions**.

Most short term forecasts use initial conditions generated with data assimilation (DA). For domains with sparse observations, including oceanic and polar regions, custom built localization techniques are required, leading to mixed efficacy and strong sensitivity of full field or anomaly initialization at different timescales [12]. DA systems are also incredibly complicated to build and maintain. To avoid this cost, some groups initialize by directly interpolating reanalysis fields like temperature and salinity onto the model grid [13]. These systems experience rapid drift from their initial conditions due to differences from the model used to fill sparse observations in the reanalysis and the forecast model, degrading predictive skill. Other studies have improved predictive skill by using initial conditions derived from a long-running ocean and sea-ice simulation forced by prescribed observed atmosphere conditions to minimize model drift [14]. Creating such initial conditions requires simulations of at least 300 yrs, leading to significant computational expense which precludes using this approach for high resolution runs.

It is clear that a new approach is needed and recent advances in generative AI methods for data assimilation are promising. Unlike traditional DA which incrementally changes a first-guess initial condition to be more consistent with subsequent observations, generative data assimilation (GenDA) directly predicts a collection of plausible initial conditions consistent with the set of available observations. In the first application of GenDA to the ocean, training on reanalysis data allowed the surface state of the ocean to be quickly 'filled in' from limited satellite data with improved reconstruction of small scale features [15]. The same authors have recently extended their work to a large ensemble from the Max Planck Institute to reconstruct 3D ocean states. This allows for generation of plausible 3D ocean states from sparse profiling floats and satellite data, but these states have yet to be tested for model prediction and this method is not subject to physical constraints. Current GenDA approaches require interpolation to regular grids which results in significant loss of information, especially in the polar regions or for models with variable resolution. The current GenDA score-based diffusion system [15] is also computationally intractable at high resolution since it operates on priors in physical space. We propose to advance the state-of-the-art in GenDA by focusing on coupled ocean and sea ice state initialization in DOE's Energy Exascale Earth System Model (E3SM) subject to essential physical constraints and train the diffusion model in latent space to dramatically increase computational efficiency [16, 17]. We will also replace the convolutional neural network (CNN) backbone of GenDA with graph based neural networks [18], which are an effective solution to this problem but have not yet been applied to GenDA for Earth system models.

2 Project Objectives

To test our core hypothesis that improved coupled ocean and sea-ice initial conditions have a strong impact on predictive skill on weeks to years timescales we will:

1. Apply GenDA to the ocean and sea ice coupled system on native unstructured grids.
2. Create workflows for integration of Surface Water and Ocean Topography (SWOT), Argo, Ice tethered profilers (ITP) and ICESat-2 data into our GenDA system that are easily portable to the American Science Cloud (AmSC).
3. Use our new initial conditions in forecasts with the E3SMv3 physical model and the corresponding E3SM-AI2 emulator.
4. Evaluate the new initial conditions against those created from conventional methodologies by leveraging existing initial conditions generated by the E3SM project.

3 Proposed Research and Methods

3.1 Task 1: Generative Data Assimilation for E3SM

In collaboration with the University of Washington and NVIDIA (see letters in Appendix 7), we treat E3SM ocean and sea ice initialization as a conditional generative data assimilation problem (Figure 1). Rather than constructing a single best estimate from sparse observations, our approach learns a probability distribution over physically consistent coupled ocean–sea-ice states using samples drawn from fully coupled E3SM simulations. We first train a variational autoencoder on these plausible priors on the native marine mesh to form a reduced dimension latent space. We then train a score-based diffusion model on the latent space by progressively increasing noise and learning the corresponding reverse-time denoising process. Training on the native marine mesh allows us to capture the full multivariate covariance structure, variability, and physical balance of the coupled system. During assimilation, the reverse-diffusion process is conditioned on observations through an observational operator, steering the decoded solution toward states that approach available measurements while remaining consistent with the distribution of physically plausible variability. Physical consistency for our prototype system is defined as enforcing mass conservation through the coupled system in constrained decoding, while physical plausibility is defined as achieving a balanced initial coupled system with minimal model drift in forward integration. Because this generative representation is learned separately from the observing system, our framework can be extended naturally to new sensors and data modalities without retraining the core model.

Ingestion of critical sparse sampled data for the ocean and sea ice system is a critical aspect of our GenDA system. Existing sea ice DA systems focus on coarse-resolution data that cannot recover critical features [19] in thin and fractured sea ice [20]. Recently, Dr. Horvat (see letter of collaboration) has developed an AI ready ICESat-2 product containing billions of samples and will help guide our integration of his product. The data are sparse locally but with high coverage, optimal for our GenDA framework. The SWOT mission [21] provides two-dimensional maps of sea surface height at unprecedented spatial scales. SWOT data has been included in a number of DA systems (e.g., [22]) providing a method to constrain high resolution ocean dynamics. To constrain the subsurface ocean we will use a balanced operator approach [23] to estimate vertical structures from surface signatures. These vertical modes will be supplemented with Argo and ITP profiling data.

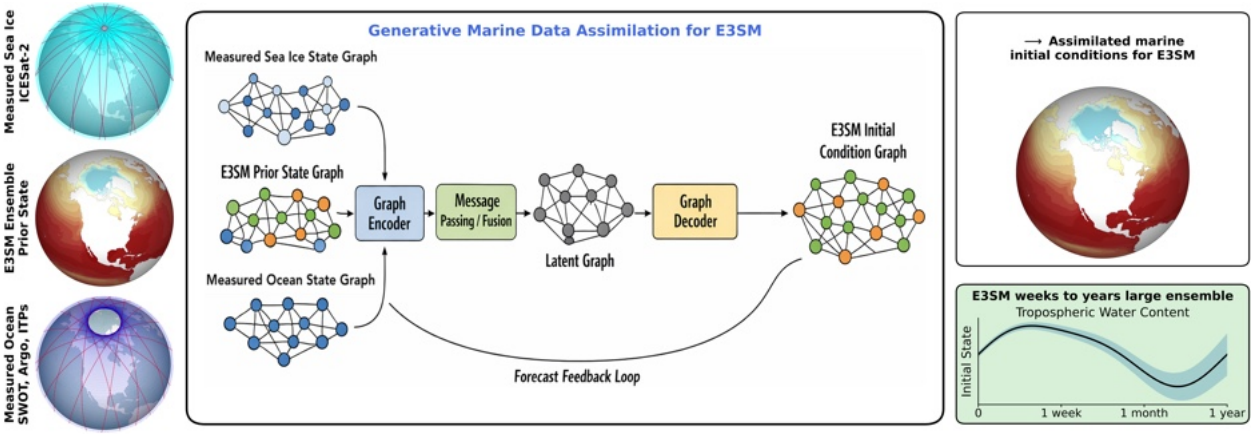


Figure 1: E3SM fusion with marine measurements to better predict U.S. tropospheric water content.

3.2 Task 2: Graph Methods

We will generalize GenDA to unstructured model meshes and multiple observational data streams using Graph Neural Networks (GNNs; Figure 1). GNNs have displayed remarkable predictive skill, rivaling the accuracy of state-of-the-art numerical weather prediction codes with vastly reduced inference time [24, 25]. These successes have since been extended to prediction of ocean properties when trained on satellite observations [26] and reanalysis products [27]. We will replace the current CNN backbone in GenDA systems with a Graph U-Net appropriate for unstructured meshes like those used in E3SM. GNNs are able to learn generalizable, equivariant spatial relationships across long distances, improving upon purely local information in standard neural networks and circumventing the need for interpolation to a regular grid.

To map observations to the model meshes for data assimilation, we will use a separate GNN-based generative model to improve upon manual interpolation that can cause information loss, induce biases, and create artifacts. Specifically, we will train a Latent Graph Diffusion (LGD) model [28] to map sparse, high-resolution Level 3 (L3) observational data to gap-free Level 4 (L4) products on E3SM model meshes. The LGD will encode both the source (observational) and target (E3SM mesh) graphs into a latent space, where diffusion is performed before decoding. We will train the LGD model on synthetic observations created by sampling E3SM fields at L3 observation (e.g., ICESat-2, SWOT) locations and using these to reconstruct full L4 fields on the E3SM mesh conditioned on the observations. During inference the LGD will use true L3 observations to generate L4 fields on the E3SM mesh for assimilation.

3.3 Task 3: Simulation campaign

To quantify improvements in performance of our initial conditions in coupled model predictions as a function of project tasks, we will perform ensembles of two-year-long E3SMv3 standard resolution on a subset of seasons from 1980-2018 after Task 1 and then perform a complete ensemble after completion of Task 2. To improve testing throughput, we will first perform ensemble experiments with E3SM's coupled SamudrACE emulator, using the slower physics-based version of E3SM to provide later confirmation of results as a phase I capstone. We will evaluate these new runs against companion simulations using traditional initialization approaches which are being run now by an E3SM-funded pilot study.

3.4 Task 4: Metric Development

The scientific impact of our new initial conditions will be evaluated using skill and bias metrics from coupled E3SM simulations initialized with our AI-generated marine states. Using a broad suite of observationally-derived ocean, polar, and atmospheric datasets, we will assess E3SM large ensembles from weeks to years after initialization, focusing on the upper ocean, sea ice, and troposphere. We will test whether our GenDA system improves predicted spatiotemporal structure of sea surface height, temperature, salinity, as well as sea-ice extent, area, and freeboard, especially in the placement, persistence, and large-scale organization of anomalies. Pattern fidelity will be assessed with self-organizing maps, clustering autoencoders, explainable neural networks, and classical metrics including Taylor's quadratic skill score, full-field pattern agreement, and ice-edge diagnostics [29, 30, 31]. We will also evaluate predictors of precipitation skill, including water vapor path, 700-hPa temperature, and 500-hPa geopotential height, using deterministic and probabilistic measures to identify forecasts of opportunity and water-supply-relevant skill [32, 33, 34, 35]. To distinguish useful regional gains from bulk global behavior, we will perform both a global survey and targeted U.S. water-supply evaluation and stratify cases by skill quintile to identify the regimes most improved by AI initialization.

3.5 Task 5: Integration with the American Science Cloud

We will develop and scale workflows using software already being prepared for AmSC, further expanding this valuable DOE resource. Specifically, we will use RHAPSODY [36], a current DOE middleware focused on enabling hybrid AI/HPC workflows, and its co-developed package, ROSE (RADICAL Orchestrator for Surrogate Exploration,) that targets the creation and validation of surrogate models of dynamical systems. As it is impossible to observe the entire state of the system, these tools effectively address the general problem of state estimation and imputation, which have similar analogues in other Genesis Challenge areas (e.g. chemistry and fusion). We will further build upon existing Genesis toolsets by developing a specific translator and data card for each of our ingested data sets, including but not limited to the NASA remote sensing information (e.g., ICESat-2). We will closely integrate with the Genesis ModCon and Data Teams to ensure our models are fully integrated to AmSC. A core deliverable of our Phase I effort will enable a wide range of Phase II projects that can immediately take advantage of and adapt our methodologies developed without the usual cost of re-implementation.

4 Milestones

Phase I will lay the foundation for improved predictability and understanding of sources of prediction skill on multi-week to years timescales. By the end of phase I we will

1. **Months 1-4** Adapt and demonstrate the UW generative data assimilation framework for coupled 3D ocean–sea-ice state representation on E3SM's unstructured grids. Demonstrate generation of mass conserving ensemble initial states compatible with E3SMv3 input formats. (LANL, CU, HPE)

2. **Months 4-6** Develop and validate a workflow for ingesting and integrating satellite observations (e.g., SWOT, ICESat-2) into the generative DA framework. Demonstrate end-to-end assimilation of at least one observational data stream and generation of conditioned ensemble initial states. (LANL, HPE)

3. **Months 1-4** Develop metrics for validation of GenDA system (LLNL, LANL)

4. **Months 4-8** Conduct hindcast experiments using E3SMv3 and an emulator initialized with GenDA ensemble states and baseline initialization. (LLNL, LANL)

5. **Months 7-9** Evaluate forecast performance using metrics defined in Task 4 to quantify the impact of generative initialization relative to current approaches. (LANL, LLNL)

5 Data sources and methods

Our work will rely heavily on E3SM version three standard resolution data. Our training will leverage existing pre-industrial control and historical period ensemble simulations. We will also incorporate a number of observational products. For the ocean, we will use SWOT satellite data, Argo profiling float data, and ice tethered profilers (ITPs). For sea ice, we will focus on integration of ICESat-2 data. At the core of our AI work is a latent score-based diffusion model framework. This powers the GenDA framework. As described in Task 2, we will augment this work with graph neural network approaches to extend the generative DA capability to unstructured models like E3SM. Our AI based initial condition system will be interfaced with the high performance workflow manager Rhapsody for smoother transition to the American Science Cloud and our graph based generative framework could be included in the Transformational AI Models Consortium.

6 Decision gate metrics

Our Phase I GenDA system will be deemed acceptable if it passes binary GO/NO-GO gates for verification, stress testing, and prototype operation. *Verification* will confirm that the marine GenDA workflow a) correctly ingests inputs, b) produces E3SM-ready mass conserving initial conditions, c) reproduces results, and d) reduces E3SM ocean-ice initialization time by a factor of two or more even when accounting for training time, as compared to conventional ocean and sea ice spinup times. *Stress Testing* will score the ability of the system to degrade safely under sparse or corrupt observations; a GO will require mass conserving model-usable outputs followed by stable year-length coupled runs, and explicit error messages when these conditions cannot be generated. *Prototype Operation* will determine whether the full workflow is scientifically viable and beneficial. A GO will require a) autonomous end-to-end execution of the GenDA system from data ingestion to initial condition creation to E3SM forward integration, b) reliable across four consecutive annual forecasts initialized at the start of successive season, c) discoverable and reusable outputs in AmSC, and d) annual prediction ensembles improving Task 4 upper ocean, sea-ice, and U.S. water-supply metrics above the baseline E3SM large ensemble.

When successful, this work lays the foundation for more rapid advances in weeks to years prediction. In phase II we will deepen our analysis of model skill, adding interpretable AI methods to understand sources of predictability (e.g., [37]). We will also extend our methodology to the land component, another important source of predictive skill [8]. Finally, we will modify our AI model to provide initial conditions at the km-scale resolution that is a core focus for E3SM.

The ultimate goal of this project is to provide DOE with what we see as the key missing link preventing it from providing useful predictions of environmental conditions weeks to years into the future. Such a capability would provide a distinct competitive advantage, positioning the resulting framework at the forefront of international marine data assimilation efforts.

Appendix 1. Bibliography & References Cited

References

[1] L. Olivetti and G. Messori, "Do data-driven models beat numerical models in forecasting weather extremes? a comparison of ifs hres, pangu-weather, and graphcast," *Geoscientific Model Development*, vol. 17, no. 21, pp. 7915–7962, 2024.

[2] Y. Shi, R. Hu, N. Wu, H. Zhang, X. Liu, Z. Zeng, J. Zhu, P. Han, C. Luo, H. Zhang, *et al.*, "Comparison of ai and nwp models in operational severe weather forecasting: A study on tropical cyclone predictions," *Journal of Geophysical Research: Machine Learning and Computation*, vol. 2, no. 2, p. e2024JH000481, 2025.

[3] J. T. Radford, I. Ebert-Uphoff, and J. Q. Stewart, "A comparison of ai weather prediction and numerical weather prediction models for 1–7-day precipitation forecasts," *Weather and Forecasting*, vol. 40, no. 4, pp. 561–575, 2025.

[4] C. Kent, A. A. Scaife, N. J. Dunstone, D. Smith, S. C. Hardiman, T. Dunstan, and O. Watt-Meyer, "Skilful global seasonal predictions from a machine learning weather model trained on reanalysis data," *npj Climate and Atmospheric Science*, vol. 8, no. 1, p. 314, 2025.

[5] L. Palma, A. Peraza, D. Civantos-Prieto, A. Duarte, S. Materia, Á. G. Muñoz, J. Peña-Izquierdo, L. Romero, A. Soret, and M. G. Donat, "Data-driven seasonal climate predictions via variational inference and transformers," *npj Climate and Atmospheric Science*, 2026.

[6] M. Bushuk, S. Ali, D. A. Bailey, Q. Bao, L. Batté, U. S. Bhatt, E. Blanchard-Wrigglesworth, E. Blockley, G. Cawley, J. Chi, F. Counillon, P. G. Coulombe, R. I. Cullather, F. X. Diebold, A. Dirkson, E. Exarchou, M. Göbel, W. Gregory, V. Guemas, L. Hamilton, B. He, S. Horvath, M. Ionita, J. E. Kay, E. Kim, N. Kimura, D. Kondrashov, Z. M. Labe, W. Lee, Y. J. Lee, C. Li, X. Li, Y. Lin, Y. Liu, W. Maslowski, F. Massonnet, W. N. Meier, W. J. Merryfield, H. Myint, J. C. A. Navarro, A. Petty, F. Qiao, D. Schröder, A. Schweiger, Q. Shu, M. Sigmond, M. Steelee, J. Stroeve, N. Sun, S. Tietsche, M. Tsamados, K. Wang, J. Wang, W. Wang, Y. Wang, Y. Wang, J. Williams, Q. Yang, X. Yuan, J. Zhang, and Y. Zhang, "Predicting September Arctic Sea Ice: A Multi-Model Seasonal Skill Comparison," *Bulletin of the American Meteorological Society*, Apr. 2024.

[7] Y. Chen, Z. Shen, Y. Tang, and X. Song, "Ocean data assimilation for the initialization of seasonal prediction with the community earth system model," *Ocean Modelling*, vol. 183, p. 102194, 2023.

[8] J. H. Richter, A. A. Glanville, T. King, S. Kumar, S. G. Yeager, N. A. Davis, Y. Duan, M. D. Fowler, A. Jaye, J. Edwards, *et al.*, "Quantifying sources of subseasonal prediction skill in cesm2," *npj Climate and Atmospheric Science*, vol. 7, no. 1, p. 59, 2024.

[9] M. Gervais, L. Sun, and C. Deser, "Impacts of Projected Arctic Sea Ice Loss on Daily Weather Patterns over North America," *Journal of Climate*, pp. 1065–1085, Feb. 2024.

[10] J. S. Pérez-Carrasquilla and M. J. Molina, "An earth-system-oriented view of the s2s predictability of north american weather regimes," *Artificial Intelligence for the Earth Systems*, vol. 4, no. 3, p. 240075, 2025.

[11] I. Cvijanovic, A. Simon, X. Levine, R. White, P. Ortega, M. Donat, D. D. Lucas, J. C. H. Chiang, A. Seidenglanz, D. Bojovic, A. R. Amaral, V. Lapin, F. Doblas-Reyes, and D. Petrova, "Arctic sea-ice loss drives a strong regional atmospheric response over the North Pacific and North Atlantic on decadal scales," *Communications Earth & Environment*, vol. 6, p. 154, Mar. 2025.

[12] D. M. Smith, R. Eade, and H. Pohlmann, "A comparison of full-field and anomaly initialization for seasonal to decadal climate prediction," *Climate dynamics*, vol. 41, no. 11, pp. 3325–3338, 2013.

[13] G. A. Meehl, B. Kirtman, A. A. Glanville, J. Richter, N. Rosenbloom, and S. Yeager, "Evaluating skill in predicting the interdecadal pacific oscillation in initialized decadal climate prediction hindcasts in e3smv1 and cesm1 using two different initialization methods and a small set of start years," *Climate Dynamics*, vol. 62, no. 2, pp. 1179–1190, 2024.

[14] S. G. Yeager, N. Rosenbloom, A. A. Glanville, X. Wu, I. Simpson, H. Li, M. J. Molina, K. Krumhardt, S. Mogen, K. Lindsay, *et al.*, "The seasonal-to-multiyear large ensemble (smyle) prediction system using the community earth system model version 2," *Geoscientific Model Development*, vol. 15, no. 16, pp. 6451–6493, 2022.

[15] S. A. Martin, G. Manucharyan, and P. Klein, "Generative Data Assimilation for Surface Ocean State Estimation from Multi-Modal Satellite Observations," Mar. 2025.

[16] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 10684–10695, 2022.

[17] Y. Qu, J. Nathaniel, S. Li, and P. Gentine, "Deep generative data assimilation in multimodal setting," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 449–459, 2024.

[18] F. Scarselli, M. Gori, A. C. Tsoi, M. Hagenbuchner, and G. Monfardini, "The graph neural network model," *IEEE transactions on neural networks*, vol. 20, no. 1, pp. 61–80, 2008.

[19] K. Hartig, E. Tziperman, and C. P. Loughner, "Processes contributing to north american cold air outbreaks based on air parcel trajectory analysis," *Journal of Climate*, vol. 36, no. 3, pp. 931–943, 2023.

[20] C. Horvat, E. Buckley, and M. Stewart, "Sea ice concentration estimates from icesat-2 linear ice fraction – part 2: Gridded data comparison and bias estimation," *The Cryosphere*, vol. 19, no. 10, pp. 4819–4833, 2025.

[21] R. Morrow, L.-L. Fu, F. Ardhuin, M. Benkiran, B. Chapron, E. Cosme, F. d'Ovidio, J. T. Farrar, S. T. Gille, G. Lapeyre, P.-Y. Le Traon, A. Pascual, A. Ponte, B. Qiu, N. Rascle, and C. Ubelmann, "Global observations of fine-scale ocean surface topography with the surface water and ocean topography (swot) mission," *Frontiers in Marine Science*, vol. 6, p. 232, 2019.

[22] E. Fouchet, M. Benkiran, P.-Y. Le Traon, and E. Remy, "Comparison of a global high-resolution ocean data assimilation system with swot observations," *Frontiers in Marine Science*, vol. 12, p. 1563934, 2025.

[23] A. T. Weaver, C. Deltel, É. Machu, S. Ricci, and N. Daget, "A multivariate balance operator for variational ocean data assimilation," *Quarterly Journal of the Royal Meteorological Society: A journal of the atmospheric sciences, applied meteorology and physical oceanography*, vol. 131, no. 613, pp. 3605–3625, 2005.

[24] R. Keisler, "Forecasting Global Weather with Graph Neural Networks," Feb. 2022. arXiv:2202.07575 [physics].

[25] R. Lam, A. Sanchez-Gonzalez, M. Willson, P. Wirnsberger, M. Fortunato, F. Alet, S. Ravuri, T. Ewalds, Z. Eaton-Rosen, W. Hu, A. Merose, S. Hoyer, G. Holland, O. Vinyals, J. Stott, A. Pritzel, S. Mohamed, and P. Battaglia, "Learning skillful medium-range global weather forecasting," *Science*, vol. 382, pp. 1416–1421, Dec. 2023.

[26] G. A. Cuervo-Londoño, J. Sánchez, and Rodríguez-Santana, "Forecasting Sea Surface Temperature from Satellite Images with Graph Neural Networks," in *Computer Analysis of Images and Patterns*, (Cham), pp. 329–339, Springer Nature Switzerland, 2026.

[27] D. Holmberg, E. Clementi, and T. Roos, "Regional Ocean Forecasting with Hierarchical Graph Neural Networks," Nov. 2024. arXiv:2410.11807 [physics].

[28] C. Zhou, X. Wang, and M. Zhang, "Unifying Generation and Prediction on Graphs with Latent Graph Diffusion," in *Advances in Neural Information Processing Systems* (A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, eds.), vol. 37, pp. 61963–61999, Curran Associates, Inc., 2024.

[29] K. E. Taylor, "Summarizing multiple aspects of model performance in a single diagram," *Journal of Geophysical Research: Atmospheres*, vol. 106, no. D7, pp. 7183–7192, 2001.

[30] H. F. Goessling, S. Tietsche, J. J. Day, E. Hawkins, and T. Jung, "Predictability of the arctic sea ice edge," *Geophysical Research Letters*, vol. 43, no. 4, pp. 1642–1650, 2016.

[31] J. Koch, M. C. Demirel, and S. Stisen, "The SPAtial EFficiency metric (SPAEF): Multiple-component evaluation of spatial patterns for optimization of hydrological models," *Geoscientific Model Development*, vol. 11, no. 5, pp. 1873–1886, 2018.

[32] M. A. Schroers and E. Martin, "Verification of atmospheric variables in S2S project models during extended extreme precipitation events," *Weather and Forecasting*, vol. 41, no. 1, pp. 3–21, 2026.

[33] Z. Zhang, M. J. DeFlorio, L. Delle Monache, A. C. Subramanian, F. M. Ralph, D. E. Waliser, M. Zheng, B. Guan, A. Goodman, A. M. Molod, F. Vitart, A. Kumar, and H. Lin, "Multi-model subseasonal prediction skill assessment of water vapor transport associated with atmospheric rivers over the western u.s.," *Journal of Geophysical Research: Atmospheres*, vol. 128, no. 7, p. e2022JD037608, 2023.

[34] P. Rivoire, O. Martius, P. Naveau, and A. Tuel, "Assessment of subseasonal-to-seasonal (S2S) ensemble extreme precipitation forecast skill over europe," *Natural Hazards and Earth System Sciences*, vol. 23, no. 8, pp. 2857–2871, 2023.

[35] N. L. Goldenson, L.-Y. R. Leung, C. M. Bitz, and E. Blanchard-Wrigglesworth, "Influence of atmospheric rivers on mountain snowpack in the western united states," *Journal of Climate*, vol. 31, no. 24, pp. 9921–9940, 2018.

[36] A. Alsaadi, M. Hooten, M. Goliyad, A. Merzky, A. Shao, M. Titov, T. Wang, Y. Chen, M. Kalantzi, K. Lee, A. Park, I. Pimpalkhare, N. Radcliffe, C. Wahl, P. Mendygral, M. Turilli, and S. Jha, "Rhapsody: Execution of hybrid ai-hpc workflows at scale," 2025.

[37] T. MacMillan and N. T. Ouellette, "Towards mechanistic understanding in a data-driven weather model: internal activations reveal interpretable physical features," *arXiv preprint arXiv:2512.24440*, 2025.

Appendix 2. Facilities & Other Resources

At Los Alamos National Laboratory, staff will have access to a large compute allocation (50 million CPU hours) on Chicoma, a HPE Cray EX computer. Chicoma has a large CPU based partition and a smaller 128 node NVIDIA A100 partition. LANL staff will also have access to the Darwin cluster, which is a heterogeneous cluster with many different GPU architectures.

The team will have access to LLNL's Tuolumne supercomputer, a smaller version of El Capitan (which is currently the world's fastest supercomputer). Tuolumne features 1,152 nodes which each contain 4 AMD MI300A GPUs. ACE already run on Tuolumne, which will also be an ideal architecture for the AI training planned for this project. Tuolumne time is not free, so we have budgeted $5K for computing access.

Alpine is the University of Colorado Boulder Research Computing's third-generation high performance computing (HPC) cluster. Alpine is a heterogeneous compute cluster currently offering 456 compute nodes and a total of 28,144 cores. Alpine can be securely accessed anywhere, anytime using Open OnDemand or ssh connectivity to the CURC system. All nodes are available to all users.

We will also leverage our connections to Office of Science program managers to ask for NERSC Mission Science computing time via an ERCAP application. Though most hours are allocated in December, the ERCAP pathway is open year round. NERSC's Perlmutter machine contains 1792 nodes with 4 NVIDIA A100 GPUs each and 3072 nodes with 2 AMD EPYC 7763 CPUs each.

Appendix 3. Equipment

No specialized equipment is required.

Appendix 5. Synergistic Activities

Luke Van Roekel

1. E3SM ocean group leader responsible for design and delivery of the new ocean component.
2. Founding member of the E3SM Seasonal to Decadal prediction group.

Andrew Roberts

1. E3SM polar group leader responsible for sea ice and land ice components and their use in fully coupled model investigations.

Katrina Bennett

1. Works within the Genesis project on data sharing.

Trevor Hillebrand

1. Developed graph neural networks for the MALI Greenland Ice Sheet model's unstructured mesh in E3SM.

Olawale Ikuyajolu

1. Founding member of the E3SM Artificial Intelligence group.

Peter Caldwell

1. Caldwell is the PI for E3SM, which is DOE's Earth system model. Caldwell has expertise and authority within the E3SM project that will be critical for of the success of this new project.

Emily de Jong

1. de Jong has a LLNL Lawrence Fellowship focused on developing an AI microphysics model and inferring 3d cloud structure from top-of-atmosphere satellite radiances. The techniques she has learned in this project will be very useful in the current project.

Christopher Horvat

1. Investigator in the Scale-Aware Sea Ice Project including the application of AI to interpreting ICESat-2 retrievals.

Andrew Shao

1. Developer of AI workflow tools SmartSim which provides interfaces from physics model codes to python AI libraries and ROSE to target efficient creation of surrogate models.

Ian Grooms

1. Charter member of the CLIVAR eddy energy team and lead developer of the GCM model filtering package.

Appendix 6. Transparency of Foreign Connections

National laboratories are not required to submit Transparency of Foreign Connections disclosures for this application in accordance with the NOFO.

Disclosure for Brown University